# Faculty of Medicine, General Sir John Kotelawala Defence University

## Application for Ethics Review (Part I) - Basic Information

| | | | |
|---|---|---|---|
| For official use | | Date Received: | |
| Application No: | | Date Informed: | |
| REC Meeting Date: | | Decision: | |
| Reviewer I: | | Decision: | |
| Reviewer II: | | Decision: | |
| Reviewer (ICF): | | | |

### 1. Title of Project

Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

### 2. Investigators & Supervisors

*Applications from investigators based overseas will only be considered if the project is done in collaboration with investigators based in institutions in Sri Lanka who take equal responsibility for the conduct of the study and who will appear as co-authors in any publication arising out of the study.*

| Title, Name and Designation of Investigators and Supervisors | Role |
|---|---|
| KH Viraj Sameera, PhD Candidate, Department of Computer Science, Faculty of Applied Sciences, University of Sri Jayewardenepura (USJ) | Principal Investigator |
| Dr. WMKS Ilmini, Department of Computer Science, Faculty of Applied Sciences, University of Sri Jayewardenepura | Principal Supervisor |
| Dr. WGCW Kumara, Department of Computer Engineering, Faculty of Engineering, University of Sri Jayewardenepura | Co-Supervisor |
| Dr. AM Deepal K Attanayake, Consultant Neurosurgeon, Neurosurgery Wards 9, 10 & 12, National Hospital of Sri Lanka (NHSL), Colombo 10 | Co-Supervisor (Clinical) / Clinical Investigator |
| Dr. IHS Kumarasinghe, MBBS, D Path, MD (Histopathology), Consultant Histopathologist / Senior Lecturer in Pathology, Department of Paraclinical Sciences, Faculty of Medicine, General Sir John Kotelawala Defence University | Co-Investigator (KDU) |

**Please note that signed and dated short curriculum vitae of all investigators and supervisors must be attached to the application.*

### 3. Contact Details of the Principal Investigator

| | |
|---|---|
| Address: | 39/3, Hills Street, Dehiwala |
| Telephone numbers: | 0775816789 |
| Fax number: | - |

| | |
|---|---|
| Email address: | virajofficialinfo@gmail.com |

**4. Funding**

| Name and Address of Funding Source(s) | Amount |
|---|---|
| Self-funded (PhD candidate). No external sponsor or commercial funding. | N/A |

**5. Proposed starting and ending dates: *‡**

Start Date: 01.04.2027 End Date: 30.09.2029

**From initial recruitment of participants until completion of all data collection.*

*‡Retrospective approval will not be given for projects already started or completed.*

No part of this study involving patient data or human participants has started. Data collection will begin only after approval from this committee and from the ethics review committee of each participating hospital.

**6. Has ethics approval for this study been requested earlier from KDU/REC or another similar committee?**

☐ Yes ☒ No

*If yes, give details (names of committees and outcome of review)*

Not previously requested. In line with the study protocol, approval will also be obtained from the Ethics Review Committee of each participating hospital (including the National Hospital of Sri Lanka) before any data are accessed at that site, and copies of those approvals will be forwarded to this committee.

*Please note that for studies sponsored by foreign funding agencies or sponsors ethics review and approval is required from the country of the funding agency or the sponsor.*

**7. Scientific Review**

Has this research proposal been subjected to scientific review by any other committee?

☒ Yes ☐ No

*If yes, give details (names of committees and outcome of review)*

Name of the committee: Faculty of Graduate Studies (FGS), University of Sri Jayewardenepura. The full PhD research proposal was submitted to the FGS on 01.10.2026 as part of PhD registration, endorsed by the supervisory panel named above. Outcome of review: awaited; the FGS decision will be forwarded to this committee as soon as it is received.

**8. Clinical Trials**

**8.1 What phase clinical trial is being conducted?**

☐ Phase I ☐ Phase II ☐ Phase III ☐ Phase IV (Post Marketing) ☒ Other

If OTHER specify: Not a clinical trial. This is a multi-centre observational study that uses routinely collected, de-identified brain-tumour data (MRI scans, digitised biopsy slides, clinical, pathology and operative reports and follow-up records) to develop and evaluate an

AI decision-support framework, together with a clinician evaluation (reader study, questionnaires and interviews). No intervention is given to patients, no additional tests, samples or visits are required, and AI outputs are not used to manage any patient during the study.

**8.2 Is it a multi-centre trial?**

☒ Yes ☐ No (multi-centre observational study, not a trial)

*If yes, list the other trial sites*

Sites: Neurosurgery Units (Wards 9, 10 & 12), National Hospital of Sri Lanka, Colombo; with the radiology and histopathology departments that hold the corresponding MRI scans and biopsy slides. Other tertiary neurosurgical units in Sri Lanka may join; each will take part only after its own ethics approval and written institutional permission. At the time of submission the confirmed site is the National Hospital of Sri Lanka; any additional site will be notified to this committee as an amendment before data collection begins there.

*Please attach ethics approval from the sponsoring country or country of the overseas principal investigator (if any). Not applicable: no foreign sponsor and no overseas investigator.*

**8.3 Is the clinical trial registered with a clinical trials registry?**

☐ Yes ☒ No

*If yes, give details (name of register and registration number)*

Not applicable (not a clinical trial).

**8.4 Data Safety Monitoring Board (only if available)**

| Name and Designation of Members | Role |
|---|---|
| Not applicable | Not applicable |

*Please attach the curriculum vitae of all members of the DSMB. Not applicable.*

**8.5 Details of Indemnity and Insurance coverage for participants, investigators and ethics committee**

Not applicable. The study involves no intervention and no physical risk to participants.

**8.6 Evidence of GCP training of ALL investigators - Attached / not attached**

Not applicable - not attached (not a clinical trial).

**9. Conflict of Interest**

**9.1 Do you believe this project has a Conflict of Interest?**

| | |
|---|---|
| Commercially | No |
| Financially | No |
| Intellectually | No |
| Other (explain) | No |

**9.2 Does any member of the research team have any affiliation with the provider(s) of funding/support, or a financial interest in the outcome of the research?**

☐ Yes ☒ No

If yes, please explain: Not applicable.

**9.3 If there is a duality of interest identified above describe the interest and state whether it constitutes a potential conflict of interest.**

Not applicable. No duality of interest or potential conflict of interest has been identified among the research team.

**9.4 Declaration of Conflict of Interest and Consent by Investigators**

We, the undersigned investigators involved in this research project, hereby:

1. Declare whether or not we have any conflict of interest (COI) related to the study (e.g., financial gain, affiliations, intellectual property, or personal relationships).

2. Confirm that we have read and understood the study protocol, agree to participate in this research project as described, and will comply with all applicable national and international ethical guidelines.

| Name of Investigator | Affiliation | COI Declared (Yes/No) | If Yes, Nature of COI | Signature | Date |
|---|---|---|---|---|---|
| KH Viraj Sameera | PhD Candidate, USJ | No | N/A | | |
| Dr. WMKS Ilmini | Principal Supervisor, USJ | No | N/A | | |
| Dr. WGCW Kumara | Co-Supervisor, USJ | No | N/A | | |
| Dr. AM Deepal K Attanayake | Co-Supervisor (Clinical), NHSL | No | N/A | | |
| Dr. IHS Kumarasinghe | Co-Investigator, KDU | No | N/A | | |

*Note: All investigators must sign individually. Unsigned forms will be considered incomplete. If a conflict of interest is declared, the REC may request further clarification or documentation.*

## Faculty of Medicine, General Sir John Kotelawala Defence University

### Application for Ethics Review (Part II) - Protocol Checklist

| For official use - Application No: | |
|---|---|

**1. Title of Protocol**

Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

**2. Name of Principal Investigator**

K.H. Viraj Sameera

**3. A List of Documents Submitted for Review**

| Title of Document | Version | Date |
|---|---|---|
| Application for Ethics Review (Part I) - Basic Information | V1 | 09.11.2026 |
| Application for Ethics Review (Part II) - Protocol Checklist | V1 | 09.11.2026 |
| Application for Ethics Review (Part III) - Document Checklist | V1 | 09.11.2026 |
| PhD Research Proposal (full study protocol) | V2 | 01.10.2026 |
| Ethics Considerations Annex (sections A1-A12) | V1 | 09.11.2026 |
| Appendix 1: Patient Information Sheet and Consent Form (English; Sinhala and Tamil translations) | V1 | 09.11.2026 |
| Appendix 2: Clinician Information Sheet and Consent Form (English; Sinhala and Tamil translations) | V1 | 09.11.2026 |
| Appendix 3: Data collection forms - case report form (English); clinician background form, reader-study case record, post-session questionnaire (System Usability Scale, NASA Task Load Index, trust) and interview guide (English, Sinhala and Tamil) | V1 | 09.11.2026 |
| Signed and dated short CVs of all investigators and supervisors | - | 09.11.2026 |

**4. Protocol Checklist**

*Section numbers refer to the attached Ethics Considerations Annex (A1-A12).*

| Please indicate the following: | Applicable (Yes/No) | Protocol Section Number |
|---|---|---|
| **Collaborative partnership** | | |
| 1. The collaborations you have established with institutions where the study is to be conducted | Yes | A2 |
| 2. The collaborations you have established with the community where the study is to be conducted | No | N/A |

| Please indicate the following: | Applicable (Yes/No) | Protocol Section Number |
|---|---|---|
| 3. The benefits to institutions, communities, and participants in your research | Yes | A3 |
| **Social value** | | |
| 1. The beneficiaries of your research and the benefit to them | Yes | A3 |
| 2. The plan for dissemination of study findings | Yes | A3 |
| **Scientific validity** | | |
| 1. The scientific importance of your study in relation to improving healthcare and/or knowledge on the subject | Yes | A1, A4 |
| 2. The justification for a replication study, if your study is a replication study | No | N/A |
| 3. How the sample size was calculated | Yes | A4 |
| **Assessment of risks/benefits** | | |
| 1. The risks to research subjects | Yes | A6 |
| 2. Benefits to research subjects | Yes | A6 |
| 3. Steps taken to minimize risks | Yes | A6, A8 |
| 4. Steps taken to enhance benefits | Yes | A6 |
| 5. Justification of the potential benefits against the risks | Yes | A6 |
| 6. Support provided to the research participants (medical, psychological and other) | Yes | A6 |
| **Consent** | | |
| 1. The procedure for initial contact of participants | Yes | A7 |
| 2. The procedure for obtaining informed consent | Yes | A7 |
| 3. The information (written/oral) provided to participants | Yes | A7, Appendix 1 & 2 |
| 4. The procedure for ensuring that subjects have understood the information provided | Yes | A7 |
| 5. The procedure for obtaining proxy consent | Yes | A7, A11 |
| 6. The procedure for withdrawing consent | Yes | A9 |
| 7. Incentives/rewards/compensation provided to participants | Yes | A7 |
| 8. The procedure for re-consenting if the research protocol changes during the course of research | Yes | A7 |
| **Confidentiality** | | |
| 1. How the data and samples will be obtained | Yes | A8 |

| Please indicate the following: | Applicable (Yes/No) | Protocol Section Number |
|---|---|---|
| 2. How long data and samples will be kept | Yes | A8 |
| 3. Justification for collection of personal identification data | Yes | A8 |
| 4. Who will have access to the personal data of the research participants | Yes | A8 |
| 5. How the confidentiality of participants will be ensured | Yes | A8 |
| 6. The procedure for data and sample storage | Yes | A8 |
| 7. The procedure for data and sample disposal | Yes | A8 |
| **Rights of the participants** | | |
| 1. Procedure for subjects to withdraw from the research at any time | Yes | A9 |
| 2. Procedure for subjects to ask questions and register complaints | Yes | A9 |
| 3. The contact person for research subjects | Yes | A9 |
| 4. Provisions for participants to be informed of results | Yes | A9 |
| 5. Provision to make the study product available to the study participants after research | No | N/A |
| **Fair participant selection** | | |
| 1. The justification for the selection of the study population | Yes | A5 |
| 2. The inclusion and exclusion criteria | Yes | A5 |
| **Responsibilities of the researcher** | | |
| 1. The provision of medical services to research participants | No | N/A |
| 2. The provisions for continuation of care after the research is completed | No | N/A |
| 3. Declaration of conflicts of interests and how the investigators plan to manage the conflicts | Yes | A10 |
| 4. The ethical / legal / social and financial issues relevant to the study | Yes | A10 |
| **Vulnerable populations** | | |
| 1. Justification for conducting the study in this population | Yes | A11 |
| **Research funded by foreign agencies / companies (not applicable - self-funded)** | | |
| 1. Justification for conducting the study in Sri Lanka | No | N/A |
| 2. Relevance of the study to Sri Lanka | No | N/A |
| 3. Post research benefits to Sri Lanka | No | N/A |
| 4. The steps taken to take into account cultural and social customs, practices, and taboos in Sri Lanka | No | N/A |

| Please indicate the following: | Applicable (Yes/No) | Protocol Section Number |
|---|---|---|
| 5. The sharing of rights to intellectual property | No | N/A |
| 6. The fate of data and biological samples including whether they will be transferred abroad and what will happen to them after the conclusion of the study | No | N/A |
| 7. How the results of research will be conveyed to relevant authorities in Sri Lanka | No | N/A |
| 8. The agreement between the sponsor/funding agency and the investigator | No | N/A |
| 9. The materials transfer agreement, if biological material is to be transferred abroad | No | N/A |
| **Community based research** | | |
| 1. The impact and relevance of the research on the community in which it is to be carried out | No | N/A |
| 2. The steps taken to consult with the concerned community during the design of the research | No | N/A |
| 3. The procedure used to obtain community consent | No | N/A |
| 4. The contribution to capacity building of the community | No | N/A |
| 5. The procedure for making available results of research to the community | No | N/A |
| **Clinical trials** | | |
| 1. Justification for withdrawing any therapy from participants to prepare them for the trial | No | N/A |
| 2. Justification for withholding standard therapy from trial participants (e.g. control group) | No | N/A |
| 3. Justification for providing care which is not the standard of care | No | N/A |
| 4. Procedure for dealing with adverse events | No | N/A |
| 5. Procedure for reporting adverse events | No | N/A |
| 6. Provisions for safety monitoring | No | N/A |
| 7. Provisions/criteria for termination of the trial | No | N/A |
| 8. Provisions for making the trial drug available to participants after the trial if found to be effective | No | N/A |

**Information Sheet (IFS) / Informed Consent Form (ICF) Check List**

Sections in the information sheets (Appendix 1 for patients, Appendix 2 for clinicians) where each item is dealt with:

| Item | Section in IFS/ICF |
|---|---|
| 1. Purpose of the study | Section 1 |
| 2. Voluntary participation | Section 2 |
| 3. Duration, procedures of the study and participant's responsibilities | Section 3 |
| 4. Potential benefits | Section 4 |
| 5. Risks, hazards and discomforts | Section 5 |
| 6. Reimbursements | Section 6 |
| 7. Confidentiality | Section 7 |
| 8. Termination of study participation | Section 8 |

**Are the investigator's qualifications and experience appropriate to conduct the study?**

☒ Yes ☐ No

| To be filled by REC Office: | |
|---|---|
| Processing charge of Rs 1000/- / Rs 2000/- | Receipt no: |
| Serial No: REC ..... / ..... | Signature REC Administrative Assistant |

## Faculty of Medicine, General Sir John Kotelawala Defence University

### Application for Ethics Review (Part III) - Document Checklist

| | |
|---|---|
| For official use - Application No: | |

**Application Checklist**

I declare that I have attached the following documents (Please tick the check box and confirm). If any of the following sections are not relevant, please indicate N/A in the given space.

| Document | Attached |
|---|---|
| 1. Application Form: Part I [1 copy] | ☒ |
| 2. Application Form: Part II [1 copy] | ☒ |
| 3. The complete research protocol including a section on ethics considerations [1 copy] (PhD Research Proposal + Ethics Considerations Annex) | ☒ |
| 4. Information sheet for research participants (Should be provided in all three languages - Sinhala, Tamil, and English - if the participant is being interviewed or is filling up the form) [1 copy each] - patients and clinicians | ☒ |
| 5. Consent forms (Should be provided in all three languages: Sinhala, Tamil, and English) [1 copy each] - patients and clinicians | ☒ |
| 6. Data collection booklets/forms/questionnaires (Should be provided in all three languages - Sinhala, Tamil, and English) [1 copy] - clinician forms 3B-3E in all three languages; case report form 3A in English only, as it is completed by hospital staff from English-language medical records and is never given to patients | ☒ |
| 7. Clinical Trials Contract (required for clinical trials) | N/A |
| 8. Materials Transfer Agreement (required for all research involving transfer of biological samples abroad) | N/A |
| 9. Indemnity/Insurance coverage (required for clinical trials) | N/A |
| 10. Ethics approval from sponsoring country or country of the overseas investigator (if any) | N/A |
| 11. Brief curriculum vitae of all investigators [1 copy] | ☒ |
| 12. Curriculum vitae of all DSMB members [1 copy] | N/A |
| 13. The downloaded payment slip for REC processing charges | ☒ |
| 14. Soft copies of all documents (The documents should be in a compressed folder (zip/rar)) have been submitted via email to ethicsreviewcommittee.fom@kdu.ac.lk | ☒ |

**The above documents should be handed over to Secretary, Research Ethics Committee, Faculty of Medicine, KDU. Deadline for submission is 10th of each month for a new protocol and re-submission.*

I understand that the application for ethics clearance will not be accepted unless all documents are submitted. I declare that I am not seeking approval for a study that has already commenced or has already been completed. I understand that at least two months are required for ethics review and granting ethics clearance.

| **Signature of Principal Investigator** | **Date** |
|---|---|
| | |

# Ethics Considerations Annex

Supplement to the PhD Research Proposal - Version 1

**Study:** Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

**Principal Investigator:** KH Viraj Sameera, PhD Candidate, Department of Computer Science, University of Sri Jayewardenepura

*This annex sets out the ethical aspects of the study. Section numbers A1-A12 are referenced in the Protocol Checklist (Part II). Section numbers of the form 3.x refer to the attached PhD Research Proposal.*

## A1. Study summary

Background. Sri Lanka records about 700 new brain and nervous-system tumours and 587 related deaths each year, with a five-year prevalence of about 2,500, but has no registry that links imaging and tissue findings. Artificial intelligence (AI) can analyse MRI scans and biopsy slides, but a clinically safe system must also know when it is not confident and must then refer the decision to a clinician (Proposal sections 1.1-1.2).

Aim. To develop and evaluate a multi-agent AI decision-support framework for the brain-tumour care pathway in which each AI component reports a calibrated confidence, gives a decision-support output only when that confidence is sufficient, and refers all other cases to a clinician, who keeps final authority at every step.

Objectives.

- O1. Design a multi-agent AI architecture for the brain-tumour pathway that passes uncertainty between components and keeps an auditable record of every AI output and clinician validation.
- O2. Develop and evaluate a calibrated, uncertainty-aware escalation mechanism that decides when a case must be referred to a clinician.
- O3. Develop data-efficient models for MRI and biopsy-slide analysis on small Sri Lankan datasets, including privacy-preserving federated training across hospitals without sharing raw patient data.
- O4. Evaluate the human-in-the-loop framework with clinicians: diagnostic performance, appropriateness of escalation, workload, usability, trust and safety, with surgical planning as a proof of concept.

Design. A multi-centre observational study with three components: (a) collection of routinely acquired clinical data (MRI scans, digitised biopsy slides, clinical, pathology and operative reports and follow-up records) into a de-identified research registry, retrospectively and prospectively; (b) computational development and evaluation of AI models on these data and on public datasets; and (c) evaluation by clinicians through a reader study, questionnaires and interviews. The study involves no intervention: no additional tests, scans, biopsies, samples or hospital visits are required, and AI outputs will not be used to diagnose or manage any patient during the study.

## A2. Study sites and collaborations

- Academic host: Department of Computer Science, Faculty of Applied Sciences, University of Sri Jayewardenepura, with the Department of Computer Engineering, Faculty of Engineering, USJ.
- Clinical site: Neurosurgery Units (Wards 9, 10 & 12), National Hospital of Sri Lanka, Colombo, under the clinical co-supervisor, together with the radiology and histopathology departments that hold the corresponding MRI scans and biopsy slides.
- Further tertiary neurosurgical units in Sri Lanka may join. Each site will take part only after approval from its own ethics review committee and written permission from the head of the institution, and under a data-sharing agreement.

Where federated learning is used, raw clinical, imaging and tissue data stay at each hospital and only model parameters are exchanged (Proposal section 3.10).

## A3. Social value, benefits and dissemination

Beneficiaries and benefits.

- Participating units: the first imaging- and tissue-linked brain-tumour registry in Sri Lanka, supporting follow-up, clinical audit and research, which remains with the units after the study.
- Future patients: AI support that could make diagnosis faster and more consistent, while referring uncertain cases to specialists.
- Clinicians: reduced review workload, measured directly in the study.
- Research community: an open, de-identified Sri Lankan brain-tumour benchmark (released only with the approvals described in A8), addressing the near-absence of South Asian data in public datasets; methods transferable to other low-resource settings.

Individual participants are not expected to benefit directly.

Dissemination. Findings will be reported in the PhD thesis, in at least four peer-reviewed publications and at conferences, and in a summary report to each participating hospital and its ethics committee, with a presentation to clinical staff. Results will be reported only in aggregate, with no information that could identify a patient or clinician. Negative results will be reported in the same detail as positive ones.

## A4. Scientific validity and sample size

Scientific importance. Existing medical AI is mostly evaluated on accuracy alone; it does not specify when a system should act and when it must defer to a clinician, and it is trained on large foreign datasets that do not represent Sri Lankan patients or scanners. This study makes clinician oversight a measurable part of the system and evaluates it on local data (Proposal sections 1.5 and 2.12). It is not a replication study.

Sample size - patients. Patients are recruited by consecutive sampling of all eligible patients at participating units during the study period. Because the number is limited by national incidence, a formal power calculation for the patient cohort is not used; the expected cohort is several hundred patients, and fewer will have complete linked imaging, tissue and follow-up data

(Proposal section 3.5). The study is designed for this: models are pre-trained on public datasets and unlabelled local scans, and results are reported with patient-level bootstrap confidence intervals so the precision of every estimate is visible.

Sample size - clinicians. Clinicians are recruited by purposive sampling. The reader study uses a multi-reader, multi-case design in which each clinician reviews a common set of cases with and without AI support, with counterbalanced order and a washout period, which uses scarce specialist time efficiently. The final number of readers and cases will be set with a statistician before the reader study begins and notified to the committee. The indicative target is at least six clinician readers.

## A5. Study populations and selection

**Population 1: Patients (data only)**

Inclusion: patients of any age with a suspected or confirmed brain or nervous-system tumour, managed at a participating unit during the study period, with at least one clinically relevant MRI examination in usable DICOM format and enough clinical information to establish the reference diagnosis.

Exclusion: records where imaging is unusable and cannot be recovered; where the reference diagnosis cannot be established; where de-identification cannot be completed; where data-governance requirements are not met; or where the patient (or proxy) has declined or withdrawn.

Patients are not excluded only because tissue, laboratory-marker or follow-up data are missing; they contribute to the parts of the study for which data exist.

**Population 2: Clinicians (active participants)**

Inclusion: consultant neurosurgeons, radiologists, histopathologists and experienced senior trainees at participating institutions who are involved in the diagnosis or management of brain tumours and who give written informed consent.

Exclusion: members of the research team (excluded from the reader study and usability evaluation).

**Population 3: Public datasets**

Public de-identified datasets such as BraTS are used for pre-training and benchmarking under their published licences. They are not used as the sole basis for any clinical claim about Sri Lankan patients.

Justification. Brain tumours are diagnosed and treated at tertiary neurosurgical units, so these are the appropriate sites. All eligible patients are included consecutively to avoid selection bias and so that the AI is evaluated on the real mix of cases, including incomplete records.

## A6. Risks, benefits and risk minimisation

Risks to patients. There is no physical risk: no procedure, test, sample or visit is added, and treatment is decided by the treating team exactly as usual. The main risk is loss of confidentiality

or re-identification from medical images; the safeguards are described in A8. The AI system is a research decision-support tool only. It will not issue or implement any clinical decision, its outputs are not used in patient care during the study, and the surgical-planning component is an offline proof of concept that never performs or controls any part of an operation.

Risks to clinicians. The main burdens are time and possible discomfort at having their reading performance measured. Sessions are scheduled at the clinician's convenience and can be paused or stopped at any time. Individual clinicians' performance is never reported by name, is never shared with employers or colleagues, and is analysed only in aggregate.

Benefits. There is no direct benefit to individual participants. Clinicians gain early experience with AI decision support. Wider benefits are described in A3. To increase benefit, the registry remains with the participating units and summary results are fed back to them.

Justification. The study is minimal risk, adds nothing to patient care, and has substantial potential benefit for brain-tumour services in Sri Lanka.

Support. Because there is no intervention, no medical or psychological support is expected to be required. Participants may contact the Principal Investigator or the clinical co-supervisor with any concern (A9). If a participating clinician raises a patient-safety concern during the study, it will be referred immediately to the clinical co-supervisor and the treating team.

## A7. Consent

**Retrospective patient records - request for waiver of consent**

For records created during routine care before the study starts, a waiver of individual consent is requested because: (i) the data already exist and no contact with the patient is needed; (ii) data are de-identified at the hospital before research use; (iii) tracing patients is impracticable, as many will have been discharged, lost to follow-up or have died, and approaching bereaved families could cause distress; (iv) the study is minimal risk; and (v) use will take place only with written permission from the head of each institution and approval from its ethics committee.

**Prospective patients - written informed consent**

Initial contact. Patients presenting during the study period are approached by a member of their treating team, not by the Principal Investigator, at a time that does not interfere with their care.

Information and consent. The patient receives the Patient Information Sheet (Appendix 1) in Sinhala, Tamil or English as preferred, has time to read it and ask questions, and gives written consent to a trained member of the clinical team. Declining has no effect on care.

Understanding. Staff explain the study orally in the patient's language and ask the patient to describe in their own words what taking part means before consent is signed.

Proxy consent. For patients under 18 years, a parent or legal guardian gives consent, and assent is sought from children old enough to understand. For adults who temporarily or permanently lack capacity, which is common with brain tumours, consent is obtained from the legally authorised representative or next of kin. If the patient later regains capacity, their own consent is sought; if they decline, their data are withdrawn.

**Clinicians - written informed consent**

Clinicians are invited by letter or email from the Principal Investigator with the Clinician Information Sheet (Appendix 2), are given at least 48 hours to decide, have the opportunity to ask questions, and sign the consent form before taking part. The information sheet, consent form, questionnaires and interview are available in Sinhala, Tamil or English, as the clinician prefers.

Incentives. No payment, gift or other incentive is offered to any participant, and taking part costs participants nothing.

Re-consent. If the protocol changes in a way that affects participants, an amendment will be submitted to the committee, and participants will be given the revised information sheet and asked to consent again.

## A8. Confidentiality and data management

How data are obtained. MRI scans are exported from the hospital imaging system by authorised hospital staff. Existing archived biopsy slides are digitised within the hospital pathology department under its own custody procedures; glass slides are not removed from the hospital and no new tissue is taken. Clinical information is abstracted from hospital records into a case report form. No biological samples are collected.

De-identification at source. Before any data leave the hospital system: direct identifiers (name, NIC number, hospital/BHT and clinic numbers, address, telephone) are removed; DICOM headers are scrubbed; structural MRI scans are defaced; and slide labels and identifying metadata are removed. Each patient is given a study code.

Justification for personal identification data. A linkage key connecting the study code to the hospital number is kept only at the hospital, by the clinical co-supervisor or designated hospital staff. It is needed only to add follow-up information and to honour withdrawal requests. No personal identifiers are held in the research dataset.

Access. The research team and trained annotators, who sign a confidentiality agreement, access de-identified data only. Only the clinical co-supervisor and designated hospital staff can access the linkage key.

Storage. Data are stored on encrypted, password-protected servers within the participating institution or the University of Sri Jayewardenepura, with role-based access control and audit logging of all access. Data are not stored on personal devices or on external cloud services. In federated experiments raw data never leave the hospital and only model parameters are shared. Clinician consent forms are kept in a locked cabinet in the Department of Computer Science, USJ, separately from questionnaire data, which are stored under a code.

Retention and disposal. Research data are kept for five years after the end of the study, or longer if required by the participating institution, and then securely deleted; paper records are shredded. The linkage key is destroyed at the end of follow-up data collection.

Open benchmark. Any public release of a de-identified dataset will happen only after a further de-identification review and with the explicit approval of the participating hospitals and their ethics committees. No data or biological material will be transferred abroad.

Legal compliance. Data will be handled in accordance with the Personal Data Protection Act, No. 9 of 2022, of Sri Lanka, and the data-governance requirements of each participating institution.

## A9. Rights of participants

Withdrawal. Participants (or their proxies) may withdraw at any time without giving a reason, and withdrawal does not affect their medical care or employment. Patients may withdraw through their treating team or by contacting the Principal Investigator; clinicians by contacting the Principal Investigator. On withdrawal, the participant's data are removed from the registry and from further analysis. Data already included in trained models or in published aggregate results cannot be removed; this is explained in the information sheets.

Questions and complaints. Participants may contact the Principal Investigator or the clinical co-supervisor with questions, and the Secretary of the Research Ethics Committee with complaints or concerns about their rights.

| | |
|---|---|
| Principal Investigator | KH Viraj Sameera - 0775816789 - virajofficialinfo@gmail.com |
| Clinical co-supervisor | Dr. AM Deepal K Attanayake, Consultant Neurosurgeon, NHSL - +94 71 421 9049 - drdeepal@yahoo.com |
| Principal Supervisor | Dr. WMKS Ilmini, Department of Computer Science, USJ - wmksilmini@sjp.ac.lk |
| Ethics Committee | Secretary, Research Ethics Committee, Faculty of Medicine, KDU - ethicsreviewcommittee.fom@kdu.ac.lk |

Results. A plain-language summary of findings will be provided to participating units and is available to any participant on request. Individual AI results are not returned to patients, because the system is a research tool and is not used for clinical decisions.

Study product. The AI framework is a research prototype, not a certified medical device, and will not be made available to participants for clinical use after the study. The registry platform remains with the participating units.

## A10. Responsibilities of the researcher and ethical, legal and social issues

Medical care. The researchers provide no medical services within the study. Patients continue to receive standard care from their treating teams during and after the study, unaffected by participation.

Conflicts of interest. No investigator has a commercial, financial or intellectual conflict of interest (Part I, section 9). Any conflict that arises during the study will be declared to the committee immediately.

AI safety. The system is decision support only; every AI output requires validation by a qualified clinician; uncertain cases are referred to clinicians rather than acted on; and every AI output and clinician decision is recorded in an audit log.

Fairness and bias. Model performance will be reported by subgroup (tumour type, site, age group and sex) so that differences in performance are visible and are not hidden in an overall figure.

Financial. Participants incur no cost. The study is self-funded.

## A11. Vulnerable populations

Two groups that may be considered vulnerable are included: children with brain tumours, and adults whose decision-making capacity is impaired by their tumour.

Justification. Brain tumours occur in children, and impaired capacity is common in exactly the most serious cases. Excluding these groups would bias the data towards milder cases and would deny future patients in these groups the benefit of a system validated on people like them.

Protections. The study adds no procedure and carries minimal risk; proxy consent and age-appropriate assent are used (A7); and all data are de-identified at source (A8).

## A12. Intellectual property and data ownership

Clinical data remain the property of the participating hospitals. Intellectual property in software and models developed in the study is governed by the intellectual-property policy of the University of Sri Jayewardenepura. Public datasets are used under their published licences. There is no foreign sponsor, and no data or biological material will be transferred outside Sri Lanka.

## Appendix 1: Patient Information Sheet

**Study:** Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

You are invited to take part in a research study. Please read this information carefully and ask us about anything that is not clear before you decide. If you are reading this on behalf of a child or a family member who cannot decide for themselves, "you" means the person you are responsible for.

**1. Purpose of the study**

We are developing a computer system that uses artificial intelligence (AI) to help doctors read brain MRI scans and biopsy slides of brain tumours. The system is designed to say when it is not sure, so that a doctor always checks those cases. This study uses information from patients' existing scans and records to build and test the system. It is part of a PhD at the University of Sri Jayewardenepura, carried out with the Neurosurgery Units of the National Hospital of Sri Lanka.

**2. Voluntary participation**

Taking part is entirely your choice. If you decide not to take part, or later change your mind, your treatment and care will not change in any way.

**3. Duration, procedures and your responsibilities**

If you agree, copies of the MRI scans, biopsy slides and medical records made during your normal care, and later follow-up information, will be used for the study for about the next two to three years. You do not need to have any extra test, scan, blood sample, biopsy or hospital visit, and you have nothing else to do. Your doctors will decide your treatment exactly as usual; the computer system is not used to make decisions about you.

**4. Potential benefits**

There is no direct benefit to you. The study may help future patients in Sri Lanka by supporting faster and more consistent diagnosis.

**5. Risks, hazards and discomforts**

There is no physical risk. The main risk is that your information could be seen by someone who should not see it. We protect against this as described in Section 7.

**6. Reimbursements**

You will not be paid for taking part, and it will not cost you anything.

**7. Confidentiality**

Your name, NIC number, hospital number, address and telephone number are removed before your information is used, and your scans are processed so that your face cannot be recognised. Your information is given a code. Only your hospital team keeps the list linking the code to you. Data are kept on secure, password-protected computers within the hospital or university. Results are published only as group totals, and you will never be named.

**8. Termination of participation**

You may stop taking part at any time without giving a reason, by telling your doctor or the researcher. Your information will then be removed from the study. Information that has already been used in published group results or in a trained computer model cannot be removed. The researchers may also stop the study early, for example if the ethics committee asks.

**Contacts**

Principal Investigator: KH Viraj Sameera, PhD Candidate, University of Sri Jayewardenepura - 0775816789 - virajofficialinfo@gmail.com

Clinical co-supervisor: Dr. AM Deepal K Attanayake, Consultant Neurosurgeon, NHSL - +94 71 421 9049

For concerns about your rights as a participant: Secretary, Research Ethics Committee, Faculty of Medicine, KDU - ethicsreviewcommittee.fom@kdu.ac.lk

## Appendix 1: Patient Consent Form

**Study:** Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

Please initial each box:

| Statement | Initials |
|---|---|
| I have read (or had read to me) the Patient Information Sheet, Version 1, and have had the chance to ask questions. | |
| I understand that taking part is voluntary and that I can withdraw at any time without my medical care being affected. | |
| I agree that copies of my MRI scans, biopsy slides, medical records and follow-up information may be used for this study after my personal details are removed. | |
| I understand that my information will be kept confidential and that I will not be named in any report or publication. | |
| I agree to take part in this study. | |

| Name | Signature | Date |
|---|---|---|
| Patient | | |
| Parent / guardian / legal representative (if applicable) - relationship: ..... | | |
| Person taking consent | | |

Assent (children old enough to understand): "The doctor explained this study to me and I am happy for my scans to be used." Child's name / signature: ..... Date: .....

## Appendix 2: Clinician Information Sheet

**Study:** Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

You are invited to take part in a research study. Please read this information carefully and ask us about anything that is not clear before you decide. You are being invited because you are involved in the diagnosis or management of patients with brain tumours.

**1. Purpose of the study**

We are developing and evaluating an AI decision-support framework for brain-tumour diagnosis and surgical-planning support that reports its own confidence and refers uncertain cases to clinicians. We want to measure whether it helps clinicians, how it affects workload, and whether clinicians find it usable and trustworthy.

**2. Voluntary participation**

Taking part is entirely voluntary. Your decision will not affect your employment, training or relationship with any participating institution.

**3. Duration, procedures and your responsibilities**

You may be asked to: (a) review a set of de-identified cases with and without AI support, in several sessions spread over time (estimated total 4-6 hours); (b) complete short questionnaires on usability, workload and trust (about 15 minutes each time); and (c) take part in an optional interview of about 30-45 minutes, which will be audio-recorded only with your separate permission. All sessions are arranged at times convenient to you.

**4. Potential benefits**

There is no direct benefit to you, although you will gain early experience with AI decision-support tools. The findings may improve brain-tumour services in Sri Lanka.

**5. Risks, hazards and discomforts**

The main burden is your time. You may feel uncomfortable having your reading performance measured; your individual results will never be reported by name or shared with your employer or colleagues.

**6. Reimbursements**

No payment is offered, and taking part costs you nothing.

**7. Confidentiality**

Your responses are stored under a code, separately from your consent form, on secure systems at the University of Sri Jayewardenepura. Results are reported only in aggregate. Interview quotations, if used, will be anonymised. Audio recordings are deleted after transcription.

**8. Termination of participation**

You may withdraw at any time without giving a reason by contacting the Principal Investigator; your data will then be excluded from further analysis. Data already included in published aggregate results cannot be removed.

**Contacts**

Principal Investigator: KH Viraj Sameera, PhD Candidate, University of Sri Jayewardenepura - 0775816789 - virajofficialinfo@gmail.com

Clinical co-supervisor: Dr. AM Deepal K Attanayake, Consultant Neurosurgeon, NHSL - +94 71 421 9049

For concerns about your rights as a participant: Secretary, Research Ethics Committee, Faculty of Medicine, KDU - ethicsreviewcommittee.fom@kdu.ac.lk

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Please initial each box:

| Statement | Initials |
|---|---|
| I have read the Clinician Information Sheet, Version 1, and have had the chance to ask questions. | |
| I understand that taking part is voluntary and that I can withdraw at any time without giving a reason. | |
| I understand that my individual performance will not be reported by name or shared with my employer. | |
| I agree to take part in the case-review sessions and questionnaires. | |
| (Optional) I agree to take part in an interview and to it being audio-recorded. | |

| Name | Signature | Date |
|---|---|---|
| Clinician | | |
| Researcher taking consent | | |